# ORF363/COS323 Convexity & Convex Optimization Part 2 (Lec5.pdf)

Ioannis Akrotirianakis

# Topics & Logistics

- Convex analysis (continue)
- $1^{st}$ & $2^{nd}$ order characterizations of convex functions
- Global optimality conditions
- Convexity of Quadratic functions
- CVX demo

# The Convex Optimization problem (The definitions & The theorem)

- A constrained optimization problem is called Convex Optimization problem iff:

$$\min_{x \in R^n} \; f(x)$$

s.t. $\quad x \in \Omega$

$$f: R^n \to R$$

$$\Omega = \{x \in R^n \mid \; g_i(x) \leq 0 \,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r \}$$

where: $g_i: R^n \to R$ and $h_j: R^n \to R$, for all $i, j$

AND $f: R^n \to R$ is convex function,
AND all $g_i: R^n \to R$ are convex functions,
AND all $h_j: R^n \to R$, are affine functions.

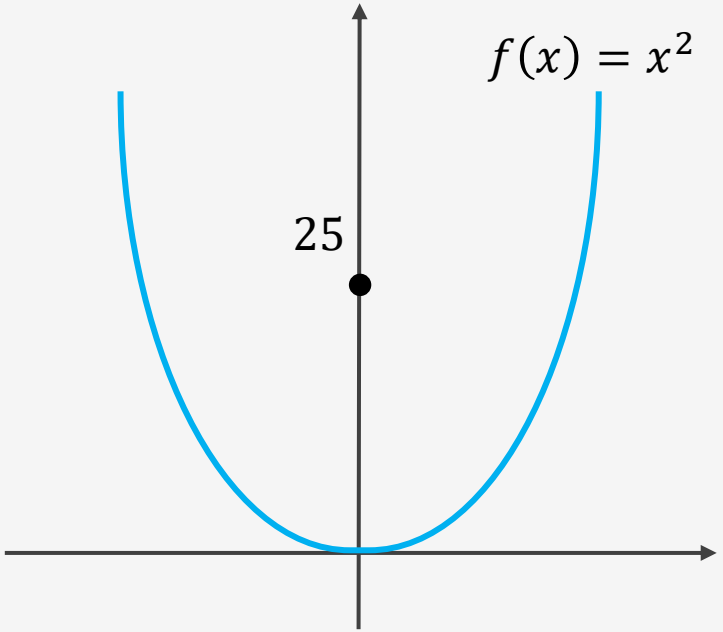
$\Rightarrow \Omega$ is a **convex set** *(because it is an intersection of convex sets)*

**Theorem:** Every local minimum of a convex optimization problem is also a global minimum

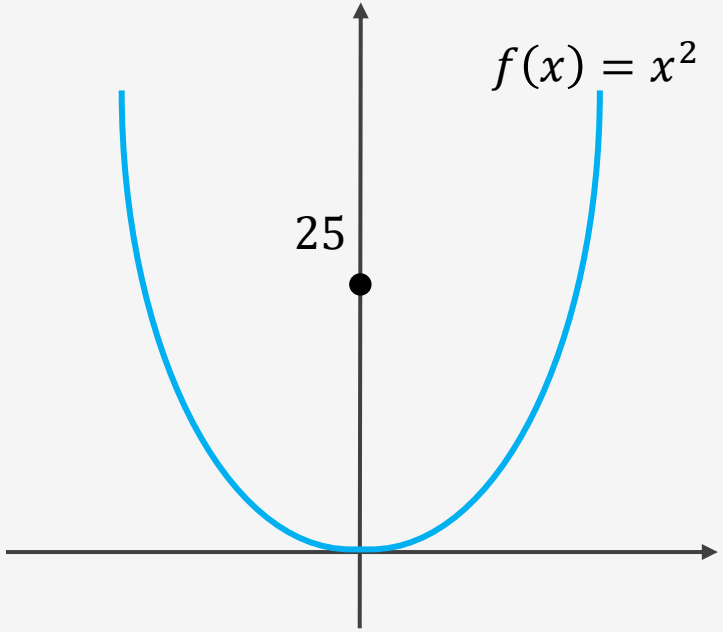
**Proof**: (see previous slides)

# Simple examples of convex feasible sets

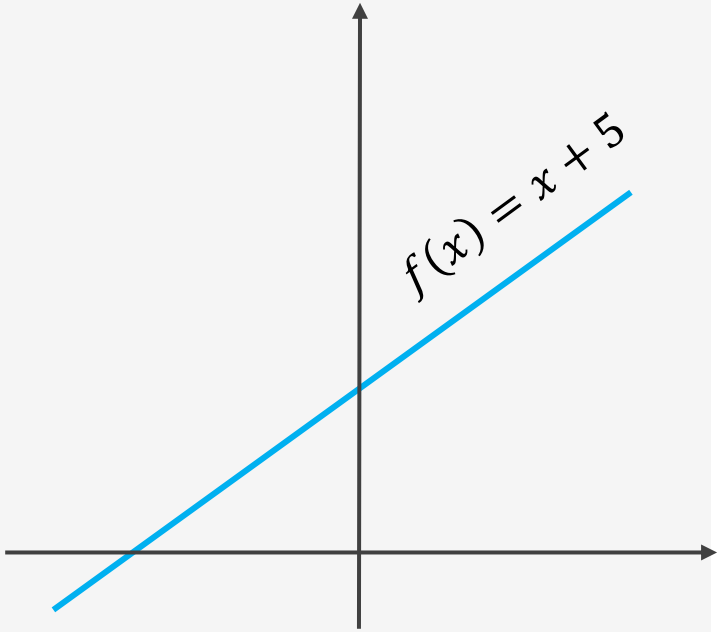
1. Find the feasible sets on the graphs/plots and decide on their convexity
2. Also, which constraints are convex?




$\Omega = \{x \in R \mid x^2 \leq 25\}$



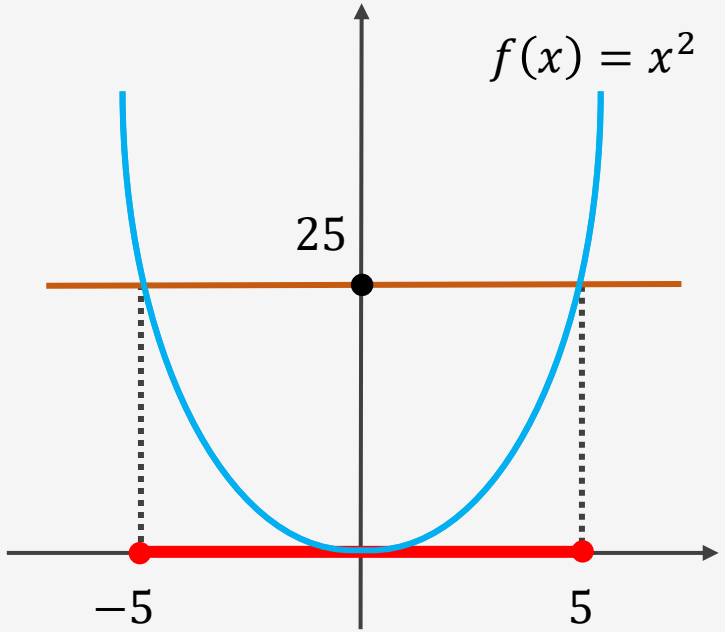

$\Omega = \{x \in R \mid x^2 = 25\}$




$\Omega = \{x \in R \mid x + 5 = 0\}$

# Simple examples of convex feasible regions

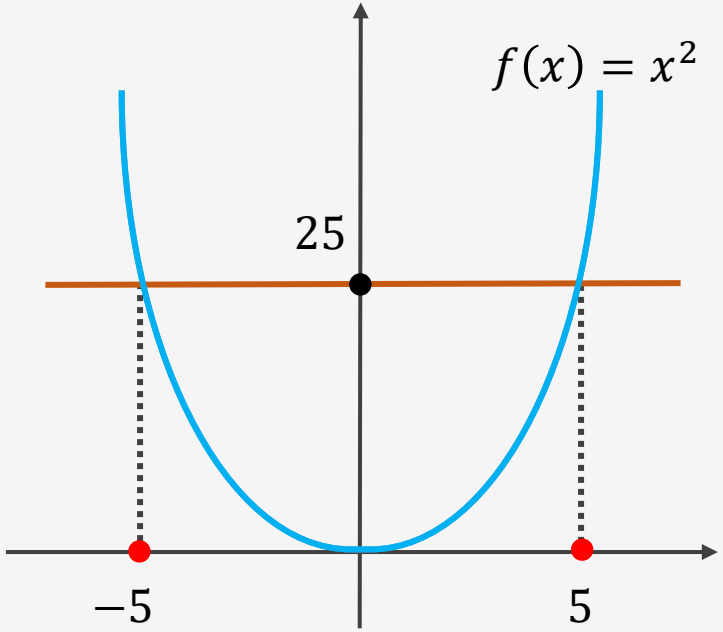
1. Find the feasible sets on the graphs/plots and decide on their convexity
2. Also, which constraints are convex?




$\Omega = \{x \in R \mid x^2 \leq 25\} = [-5,5]$

Convex constraint

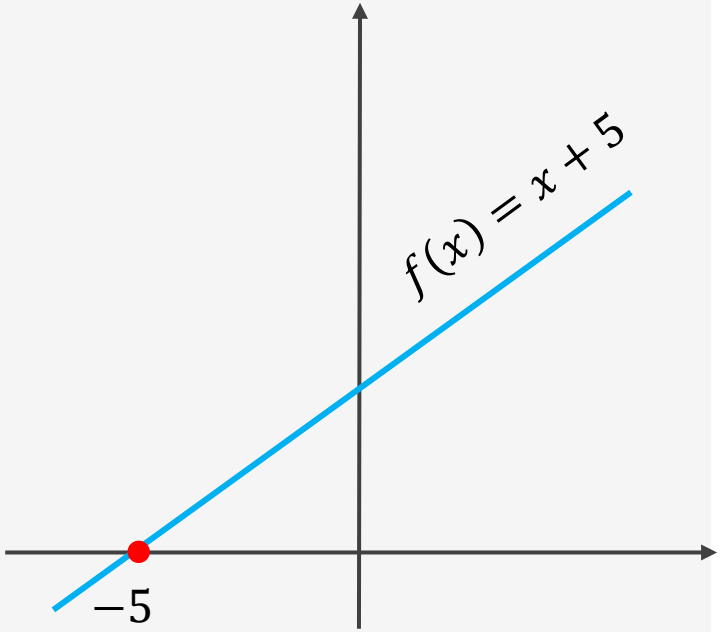
$\Omega$ is convex set




$\Omega = \{x \in R \mid x^2 = 25\} = \{-5,5\}$

Nonconvex constraint

$\Omega$ is not a convex set




$\Omega = \{x \in R \mid x + 5 = 0\} = \{-5\}$

Convex constraint

$\Omega$ is a convex set

# The Feasible Region of Convex Optimization problems

- $\Omega$ is a convex set – WHY??

$$\Omega = \{x \in R^n \mid \quad g_i(x) \leq 0\,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r\,\}$$

AND all $g_i: R^n \to R$ are convex functions,
AND all $h_j: R^n \to R$, are affine functions.

# The Feasible Region of Convex Optimization problems

- It is a convex set – WHY??

$$\Omega = \{x \in R^n \mid \quad g_i(x) \leq 0\,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r\,\}$$

AND all $g_i: R^n \to R$ are convex functions,
AND all $h_j: R^n \to R$, are affine functions.

$\Omega = \Omega_1 \cap \cdots \cap \Omega_m \cap \Omega_{m+1} \cap \cdots \cap \Omega_{m+r}$ → **<u>intersection</u>** of $m + r$ convex sets

$\Omega_i = \{x \in R^n \mid g_i(x) \leq 0\}, \qquad i = 1, \dots m$ → $m$ convex sets (because all $g_i(x)$ are convex functions)

$\Omega_{m+j} = \{x \in R^n \mid h_j(x) = 0\}, \quad j = 1, \dots, r$ → $r$ convex sets (because all $h_j(x)$ are affine functions)

**<u>Exercise:</u>** Are the following sets convex or not? If not provide a counterexample:

**1.** $\widehat{\Omega} = \{x \in R^2 \mid x_1^2 + x_2^2 \leq 1, \quad x_1 + x_2 = 0\}$ **2.** $\overline{\Omega} = \{x \in R^2 \mid x_1^2 + x_2^2 \leq 1, \quad x_1^2 + x_2^2 \geq \frac{1}{4}\}$

# 1st and 2nd order characterization of convex functions

- **Theorem**: suppose $f: R^n \to R$ is twice continuously differentiable. Then the following statements are equivalent

Definition of convexity – no derivatives needed

- **0th order:** $f$ is convex $\overbrace{(f(\lambda x + (1-\lambda)y \leq \lambda f(x) + (1-\lambda)f(y), \boxed{\forall x, y \in R^n, \lambda \in [0,1]})}$

$$\Updownarrow$$

Does it remind you anything??

- **1st order:** $f(y) \geq \overbrace{f(x) + \nabla f(x)^T(y-x)}, \boxed{\forall x, y \in R^n}$ 1st order derivatives needed

$$\Updownarrow$$

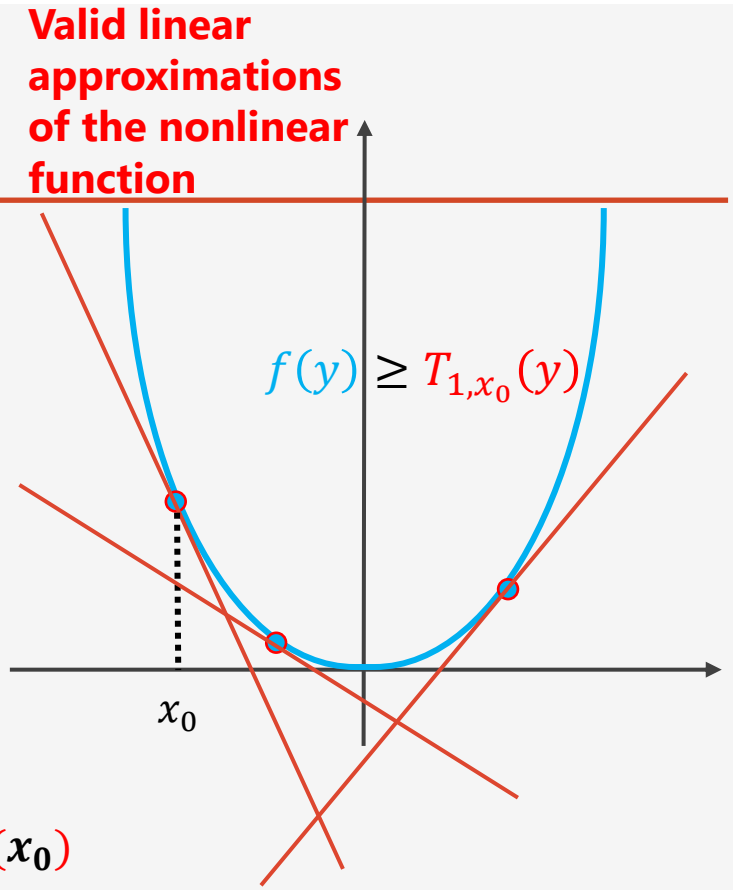
- **2nd order:** $\nabla^2 f(x) \succcurlyeq 0, \boxed{\forall x \in R^n}$ 2nd order derivatives needed

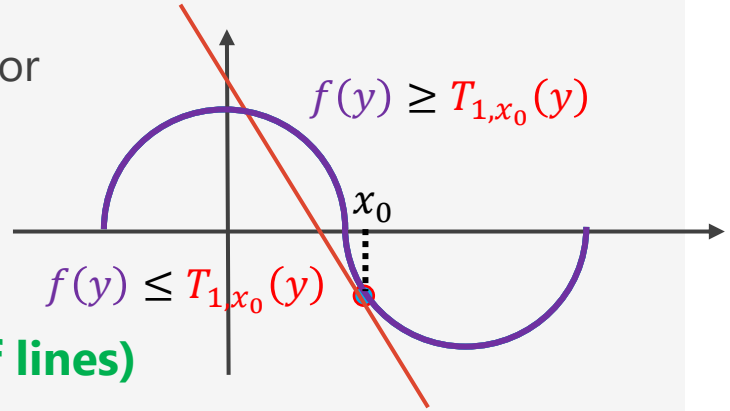
**Proof**: see pdf notes (Lec5p5)

# Intuition for the 1st order characterization

- It uses the **1st order Taylor expansion** at $x_0$
  - $T_{1,x_0}(y) = f(x_0) + \nabla f(x_0)^T (y - x_0), y \in R^n$
- It requires that the function value $f(y)$ is always $\geq$ **(above)** $T_1(y)$
  - $f(y) \geq T_{1,x_0}(y)$ ($T_{1,x_0}$ is a ***global underestimator*** of $f$)
- But $T_{1,x_0}(y)$ is **tangent** to $f$ at $x_0$ → **Where do we have $f(x_0) = T_{1,x_0}(x_0)$**
- A convex function is always above the Hyperplane defined by the 1st order Taylor expansion at every point.
  - This does not hold for **nonconvex functions**

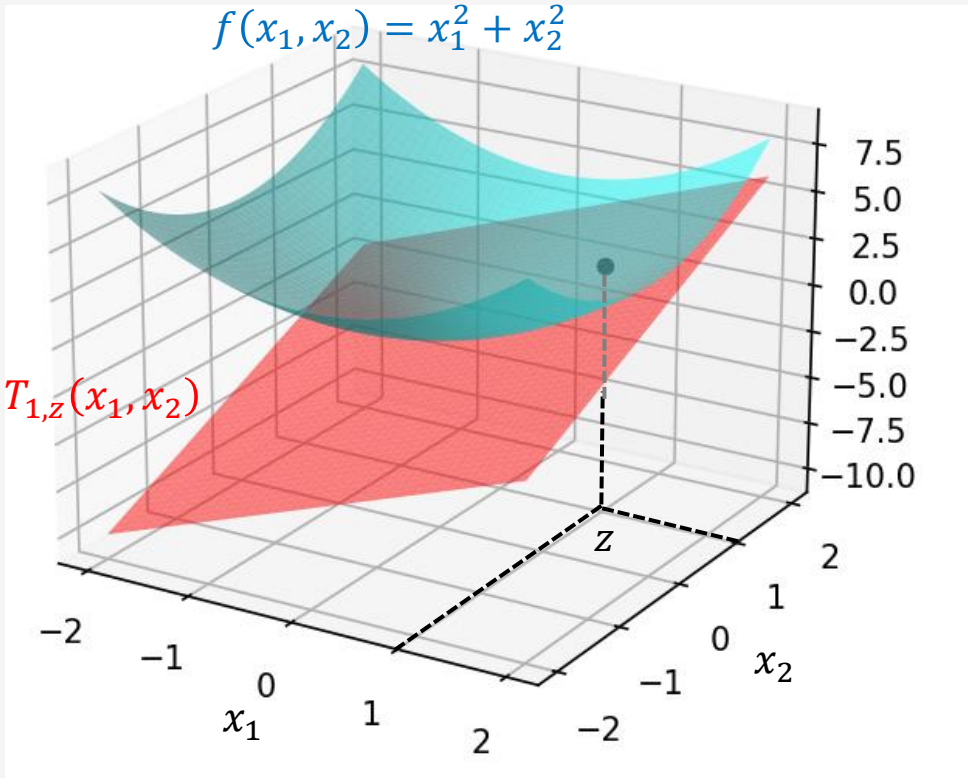
**The same intuition holds for higher dimension, $n \geq 2$ (hyperplanes instead of lines)**



# Intuition for the 1st order characterization

**The same intuition holds for higher dimension, $n \geq 2$ (hyperplanes instead of lines)**



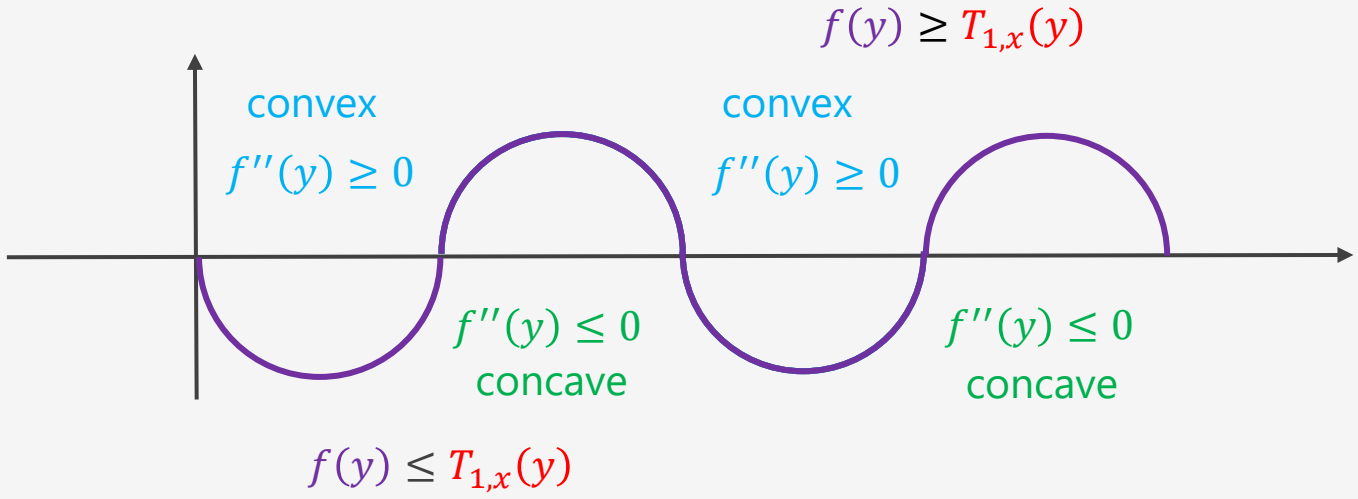

$$f(x) \geq f(z) + \nabla f(z)^T (z - x)\,, \forall x \in R^2$$

$$x = (x_1, x_2) \quad \text{and} \quad z = (1,1)$$

# Intuition for the 2nd order characterization

- From Calculus we know that $f''(x)$ expresses curvature information.

- **For $n = 1$ (one variable functions)** we know that when $f''(x) \geq 0$, the function is curving up, and when $f''(x) \leq 0$, the function is curving down.

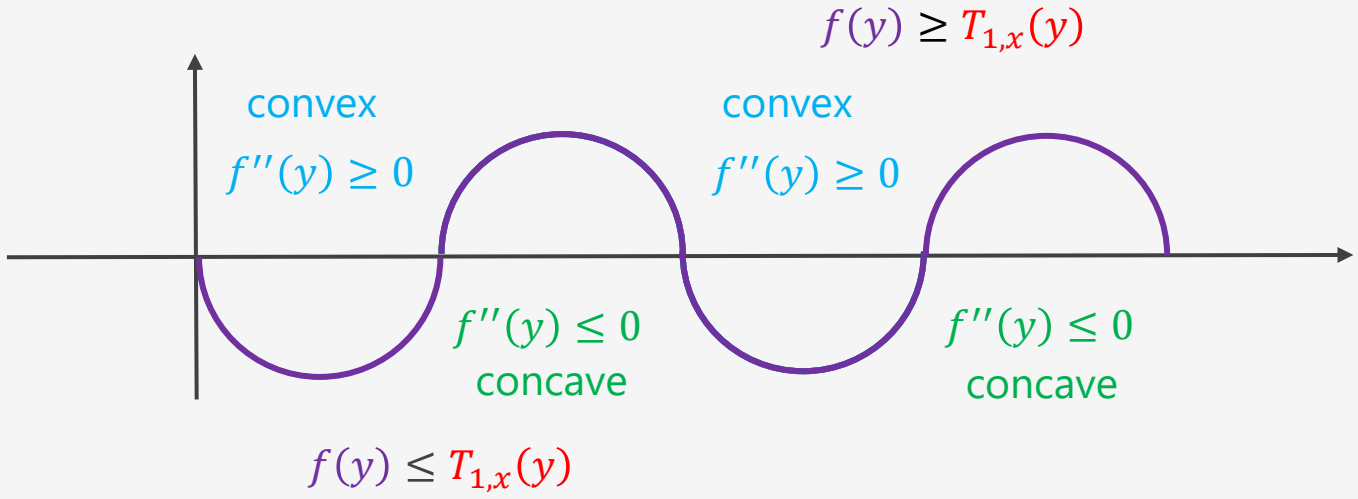



**Same intuitions for $n \geq 2$, but how could we generalize it??**

# Intuition for the 2nd order characterization

- From Calculus we know that $f''(x)$ expresses curvature information.

- **For $n = 1$ (one variable functions)** we know that when $f''(x) \geq 0$, the function is curving up, and when $f''(x) \leq 0$, the function is curving down.




**Same intuitions for $n \geq 2$, but how could we generalize it?? → $y^T \nabla^2 f(x) y \geq 0$, *or* $y^T \nabla^2 f(x) y \leq 0$, $\forall x, y \in R^n$**

# Global Optimality Condition for Minimizing Convex functions

- **Theorem**: Consider the unconstrained optimization problem and suppose $f: R^n \to R$ is a convex and differentiable function.

  $x^* \in R^n$ is a global minimum $\Leftrightarrow \nabla f(x^*) = 0$

- **Proof**: $\Rightarrow$ (what does it remind you??)

$$\min_{x \in R^n} \; f(x)$$

$$\text{s.t.} \qquad x \in \mathrm{R}^{\mathrm{n}}$$

# Global Optimality Condition for Minimizing Convex functions

- **Theorem**: Consider the *unconstrained* optimization problem and suppose $f: R^n \to R$ is a convex and differentiable function.

$$\min_{x \in R^n} \quad f(x)$$

$$\text{s.t.} \qquad x \in \mathrm{R}^{\mathrm{n}}$$

$x^* \in R^n$ is a global minimum $\Leftrightarrow \nabla f(x^*) = 0$

- **Proof**: $\Rightarrow$ This is just FONC

# Global Optimality Condition for Minimizing Convex functions

- **Theorem**: Consider the *unconstrained* optimization problem and suppose $f: R^n \to R$ is a convex and differentiable function.

$$\min_{x \in R^n} \; f(x)$$

$$\text{s.t.} \qquad x \in \mathrm{R}^{\mathrm{n}}$$

$x^* \in R^n$ is a global minimum $\Leftrightarrow \nabla f(x^*) = 0$

- **Proof**: $\Rightarrow$ This is just FONC

- $\Leftarrow$ since $f$ is convex, from the <u>1st order characterization</u>: $f(y) \geq f(x) + \nabla f(x)^T (y - x), \forall x, y \in R^n$

  - **Fixing $\boldsymbol{x} = \boldsymbol{x}^*$** yields $f(y) \geq f(x^*) + \nabla f(x^*)^T (y - x^*)$, for every $y \in R^n$

Any ideas how to proceed??

# Global Optimality Condition for Minimizing Convex functions

- **Theorem**: Consider the *unconstrained* optimization problem and suppose $f: R^n \to R$ is a convex and differentiable function.

$$\min_{x \in R^n} \; f(x)$$

$$\text{s.t.} \qquad x \in \mathrm{R}^{\mathrm{n}}$$

  $x^* \in R^n$ is a global minimum $\Leftrightarrow \nabla f(x^*) = 0$

- **Proof**: $\Rightarrow$ This is just FONC

- $\Leftarrow$ since $f$ is convex, from the <u>1st order characterization</u>: $f(y) \geq f(x) + \nabla f(x)^T (y - x), \forall x, y \in R^n$
  - **Fixing $\boldsymbol{x} = \boldsymbol{x}^*$** yields $f(y) \geq f(x^*) + \nabla f(x^*)^T (y - x^*)$, for every $y \in R^n$
  - Using the fact: $\nabla f(x^*) = 0$, we have: $\boldsymbol{f(y) \geq f(x^*)}$ **for every $\boldsymbol{y \in R^n}$**.
  - Hence $\boldsymbol{x}^*$ is a global minimum ■

# Global Optimality Condition for Minimizing Convex functions

- **Remarks**
  - 1. For convex (*unconstrained*) problems $\nabla f(x) = 0$ (FONC) is ALSO sufficient for local optimality
  - 2. But always remember, for nonconvex functions $\nabla f(x) = 0$ is **NOT** sufficient! (for example: $f(x) = x^3$)
  - 3. Also, a convex function automatically satisfies: $\nabla^2 f(x) \succcurlyeq 0$ (SONC)

# Where to use the 1st & 2nd order Characterizations

- **Usefulness** of 1st and 2nd order characterizations:
  - Usually, we use
    - The **1st order** characterization to **prove optimality conditions**
    - The **2nd order** characterization for **checking convexity** of $f$
      - Especially for <u>quadratic functions</u>

# Observations!

- Strict Convexity $\Rightarrow$ Convexity $\Rightarrow$ Quasi Convexity

# Observations!

- Strict Convexity $\Rightarrow$ Convexity $\Rightarrow$ Quasi Convexity
- Strict Convexity $\not\Leftarrow$ Convexity $\not\Leftarrow$ Quasi Convexity
  - Can you think of counterexamples??

# Observations!

- Strict Convexity $\Rightarrow$ Convexity $\Rightarrow$ Quasi Convexity

- Strict Convexity $\not\Leftarrow$ Convexity $\not\Leftarrow$ Quasi Convexity

  $f(x) = x$ $\qquad$ $f(x) = x^3$

  - Can you think of **counterexamples**

# Observations!

- Strict Convexity $\Rightarrow$ Convexity $\Rightarrow$ Quasi Convexity

- Strict Convexity $\not\Leftarrow$ Convexity $\not\Leftarrow$ Quasi Convexity

  $f(x) = x$ $\qquad$ $f(x) = x^3$

  - Can you think of **counterexamples**

- **Theorem**: Consider the *general constrained optimization problem*

$$(P1)$$

$$\boxed{\begin{array}{ll} \min\limits_{x \in R^n} & f(x) \\ \text{s.t.} & x \in \Omega \end{array}}$$

  - Suppose $\Omega$ is convex set and $f$ is **strictly** convex function.

  - **If the optimization problem has a solution, then that solution is unique**

# Remarks

- **Remark 1:** Standard convexity alone is not enough to guarantee uniqueness of a local minimum for the general constrained optimization problem (P1)
  - Think of this optimization problem:
    - how many local minima does it have??:

$$\min_{x \in R} \quad f(x) = 5$$

$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

# Remarks

- **Remark 1:** Standard convexity alone is not enough to guarantee uniqueness of a local minimum for the general constrained optimization problem (P1)
  - Think of this optimization problem:
    - how many local minima does it have??:

$$\min_{x \in R} \quad f(x) = 5$$
$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

- **Remark 2:** the optimal solution of problem (P1) may not exist, even if $f$ is strictly convex function
  - Think of this constrained optimization problem:
    - What is its optimal unconstrained solution??
    - What is the optimal constrained solutions??

$$\min_{x \in R} \quad f(x) = e^x$$
$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

# Observations

- **Remark 1:** Standard convexity alone is not enough to guarantee uniqueness of a local minimum for the general constrained optimization problem (P1)
  - Think of this optimization problem:
    - How many local minima does it have??:

**Is it a convex function?**
**How about strictly convex ??**

$$\min_{x \in R} \quad f(x) = 5$$

**Is it a convex set ??**

$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

- **Remark 2:** the optimal solution of problem (P1) may not exist, even if $f$ is strictly convex function
  - Think of this constrained optimization problem:
    - What is its optimal unconstrained solution??
    - What is the optimal constrained solutions??

**Is it a strictly convex function ??**
**Is it bounded in Ω ??**

$$\min_{x \in R} \quad f(x) = e^x$$

**Is it a convex set ??**

$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

# Observations

- **Remark 1:** Standard convexity alone is not enough to guarantee uniqueness of a local minimum for the general constrained optimization problem (P1)
  - Think of this optimization problem:
    - How many local minima does it have??:

**Is it a convex function?**
**How about strictly convex ??**

$$\min_{x \in R} \quad f(x) = 5$$

**Is it a convex set ??**

$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

- **Remark 2:** the optimal solution of problem (P1) may not exist, even if $f$ is strictly convex function
  - Think of this constrained optimization problem:
    - What is its optimal unconstrained solution??
      $x^*$ is not attained; but $f(x^*) = 0$
    - What is the optimal constrained solutions??
      $x^* = -3$ and $f(x^*) = e^{-3}$

**Is it a strictly convex function ??**
**Is it bounded in Ω ??**

$$\min_{x \in R} \quad f(x) = e^x$$

**Is it a convex set ??**

$$s.t. \quad x \in \Omega = \{x \in R \mid -3 \leq x \leq 7\}$$

# 1st and 2nd order characterization of strictly convex functions

- **Theorem**: suppose $f: R^n \to R$ is twice continuously differentiable. Then the following statements are equivalent

  - **0th order:** $f$ is **strictly** convex $\quad (f(\lambda x + (1-\lambda)y < \lambda f(x) + (1-\lambda)f(y), \forall x, y, x \neq y \in R^n, \lambda \in (0,1))$

$$\Updownarrow$$

  - **1st order:** $f(y) > f(x) + \nabla f(x)^T (x - y), \quad \forall x, y, x \neq y \in R^n$

$$\boxed{\Uparrow}$$

  - **2nd order:** $\nabla^2 f(x) \succ 0, \quad \forall x \in R^n$

# Observation

- If you are asked to prove that a function is strictly convex, you could try to show that $\nabla^2 f(x) \succ 0, \forall x \in R^n$ (i.e., use 2nd order characterization).

- Then you can also claim that the optimal solution is unique

- **Example**: consider the function $f(x) = x^4$

  - $f(x)$ is **strictly** convex, but we **cannot claim** that $\nabla^2 f(x) \succ 0, \forall x \in R$

    - For $x = 0$ we have $\nabla^2 f(x) = 0$

  - That is: 0th or 1st order characterizations for **strictly** convex functions $\not\Rightarrow$ 2nd order characterization for **strictly** convex functions

# Quadratic functions & Convexity

- Consider the <u>unconstrained</u> Quadratic Programming problem $(QP1)$:

$$\boxed{\min_{x \in R^n} \; f(x) = 0.5\, x^T Q x + b^T x + c} \qquad (QP1)$$

- **Properties**:
  - <u>Prop. 1</u>: $f(x)$ is not convex $\Leftrightarrow Q \ngeqslant 0 \Leftrightarrow f(x)$ is unbounded from below
  - <u>Prop. 2</u>: $f(x)$ is convex $\Leftrightarrow Q \succcurlyeq 0 \Leftrightarrow \nabla^2 f(x) \succcurlyeq 0, \forall x \in R^n$
  - <u>Prop. 3</u>: $f(x)$ is **strictly** convex $\Leftarrow Q \succ 0$

# Quadratic functions & Convexity

- **<u>Prop. 1</u>:** let's prove this property: If $Q \ngeq 0 \iff f$ is unbounded

- **Proof**: We will use the relation between eigenvalues and INDEFINITE matrices.

# Quadratic functions & Convexity

- **<u>Prop. 1</u>:** let's prove this property: If $Q \ngeq 0 \iff f$ is unbounded

- **Proof**: since $Q \ngeq 0$, there must be at least 1 eigenvalue $\lambda < 0$.

# Quadratic functions & Convexity

- **<u>Prop. 1</u>:** let's prove this property: If $Q \ngeq 0 \iff f$ is unbounded

- **Proof**: since $Q \ngeq 0$, there must be at least 1 eigenvalue $\lambda < 0$.

  - Let's denote by $x_\lambda$ the corresponding eigenvector. Then we have: $Qx_\lambda = \lambda x_\lambda \Rightarrow x_\lambda^T Q x_\lambda = \lambda x_\lambda^T x_\lambda = \lambda \left|\left|x_\lambda\right|\right|^2 < 0$

# Quadratic functions & Convexity

- **<u>Prop. 1</u>:** let's prove this property: If $Q \ngeq 0 \iff f$ is unbounded

- **Proof**: since $Q \ngeq 0$, there must be at least 1 eigenvalue $\lambda < 0$.

  - Let's denote by $x_\lambda$ the corresponding eigenvector. Then we have: $Qx_\lambda = \lambda x_\lambda \Rightarrow x_\lambda^T Q x_\lambda = \lambda x_\lambda^T x_\lambda = \lambda \left|\left|x_\lambda\right|\right|^2 < 0$

  - Now take a random number $\alpha > 0$ and evaluate $f$ at the point $\alpha x_\lambda$:

  - $f(\alpha x_\lambda) = 0.5\, \alpha^2 x_\lambda^T Q x_\lambda + \alpha b^T x_\lambda + c = 0.5\, \alpha^2 x_\lambda^T (\lambda x_\lambda) + \alpha b^T x_\lambda + c = 0.5\, \alpha^2 \lambda \left|\left|x_\lambda\right|\right|^2 + \alpha b^T x_\lambda + c$

# Quadratic functions & Convexity

- **<u>Prop. 1</u>:** let's prove this property: If $Q \ngeq 0 \iff f$ is unbounded

- **Proof**: since $Q \ngeq 0$, there must be at least 1 eigenvalue $\lambda < 0$.

  - Let's denote by $x_\lambda$ the corresponding eigenvector. Then we have: $Qx_\lambda = \lambda x_\lambda \Rightarrow x_\lambda^T Q x_\lambda = \lambda x_\lambda^T x_\lambda = \lambda \left|\left|x_\lambda\right|\right|^2 < 0$

  - Now take a random number $\alpha > 0$ and evaluate $f$ at the point $\alpha x_\lambda$:

  - $f(\alpha x_\lambda) = 0.5\, \alpha^2 x_\lambda^T Q x_\lambda + \alpha b^T x_\lambda + c = 0.5\, \alpha^2 x_\lambda^T (\lambda x_\lambda) + \alpha b^T x_\lambda + c = 0.5\, \alpha^2 \lambda \left|\left|x_\lambda\right|\right|^2 + \alpha b^T x_\lambda + c$

  - Take the limit: $\lim_{a \to \infty} f(\alpha x_\lambda) = -\infty$ (WHY??).

$\blacksquare$

**Exercise**: prove Prop. 2 and Pro. 3

# Quadratic functions & Convexity

- Using **Prop. 3** we can compute the **unique optimal value** of the <u>unconstrained</u> Quadratic Programming problem
  - Indeed, **since $Q \succ 0$**, we know $Q$ should be **invertible**
    - WHY?? (<u>Hint</u>: think of the eigenvalues of $Q$ and that $Qx = 0 \Rightarrow x = 0$)
  - **FONC**: $\nabla f(x^*) = 0 \Rightarrow Qx^* + b = 0 \Rightarrow x^* = -Q^{-1}b$
  - Using **SOSC** (i.e., $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) = Q \succ 0$) we can conclude that $x^* = -Q^{-1}b$ is the <u>unique global minimum</u>

# Optimality Conditions for Constrained Convex Optimization

$$\min_{x \in R^n} \; f(x)$$

s.t. $\quad x \in \Omega$

- **Theorem:** Consider the constrained Convex Optimization problem, where $f: R^n \to R$ is convex function and $\Omega \subseteq R^n$ is convex set. Then:

$$\boxed{x^* \in \Omega \text{ optimal} \quad \Leftrightarrow \quad x^* \in \Omega \quad \text{and} \quad \nabla f(x^*)^T (y - x^*) \geq 0, \forall y \in \Omega}$$

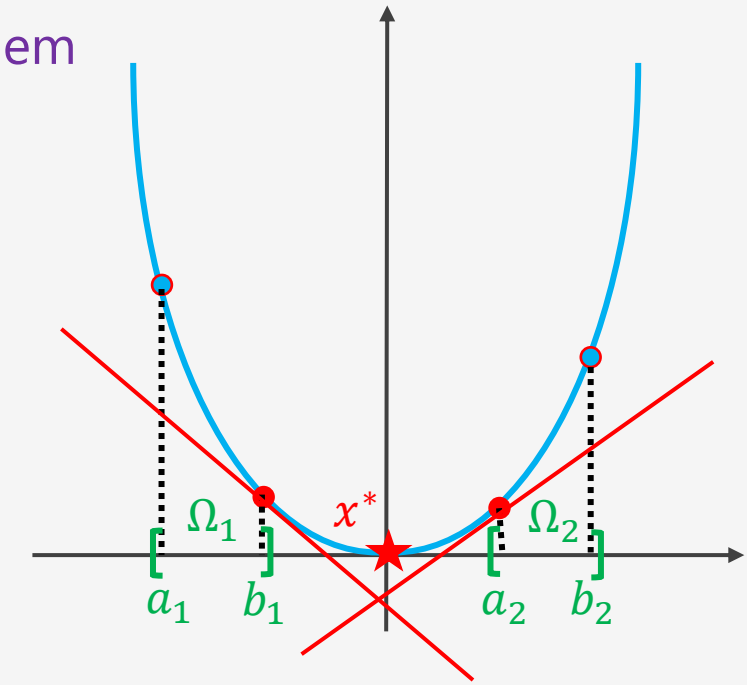
- **Remark:** when you have to solve a constrained optimization problem (convex or not) **DO NOT use FONC**

  - Remember: FONC is for unconstrained optimization, only!

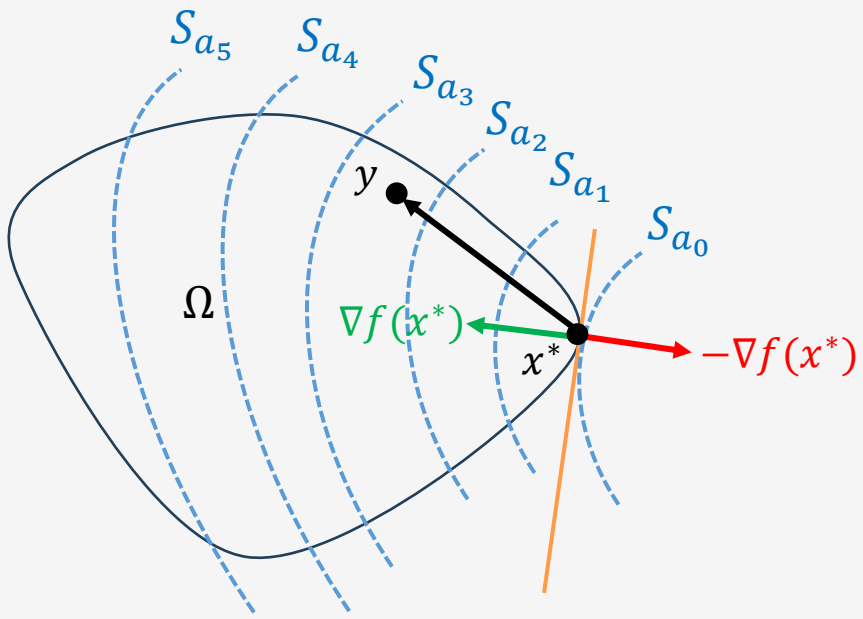
    ★ Unconstrained optimum of $f$ $\quad (\nabla f(x^*) = 0)$

    ● Constrained optimum of $f$ $\quad$ (in $\Omega_1 = [a_1, b_1]$ or in $\Omega_2 = [a_2, b_2]$)

    $\nabla f(b_1) \neq 0 \qquad \nabla f(a_2) \neq 0$



# Optimality Conditions for Constrained Convex Optimization




$$\min_{x \in R^n} \; f(x)$$

s.t. $\quad x \in \Omega$

**Geometric interpretation:**

Level sets: $S_{a_j} = \{x \in \Omega \mid f(x) = a_j\}, j = 0,1,\dots,5$

$$a_5 > a_4 > \cdots > a_0$$

$x^*$: constrained minimum $(x^* \in \Omega)$
$y$ : any feasible point $(y \in \Omega)$

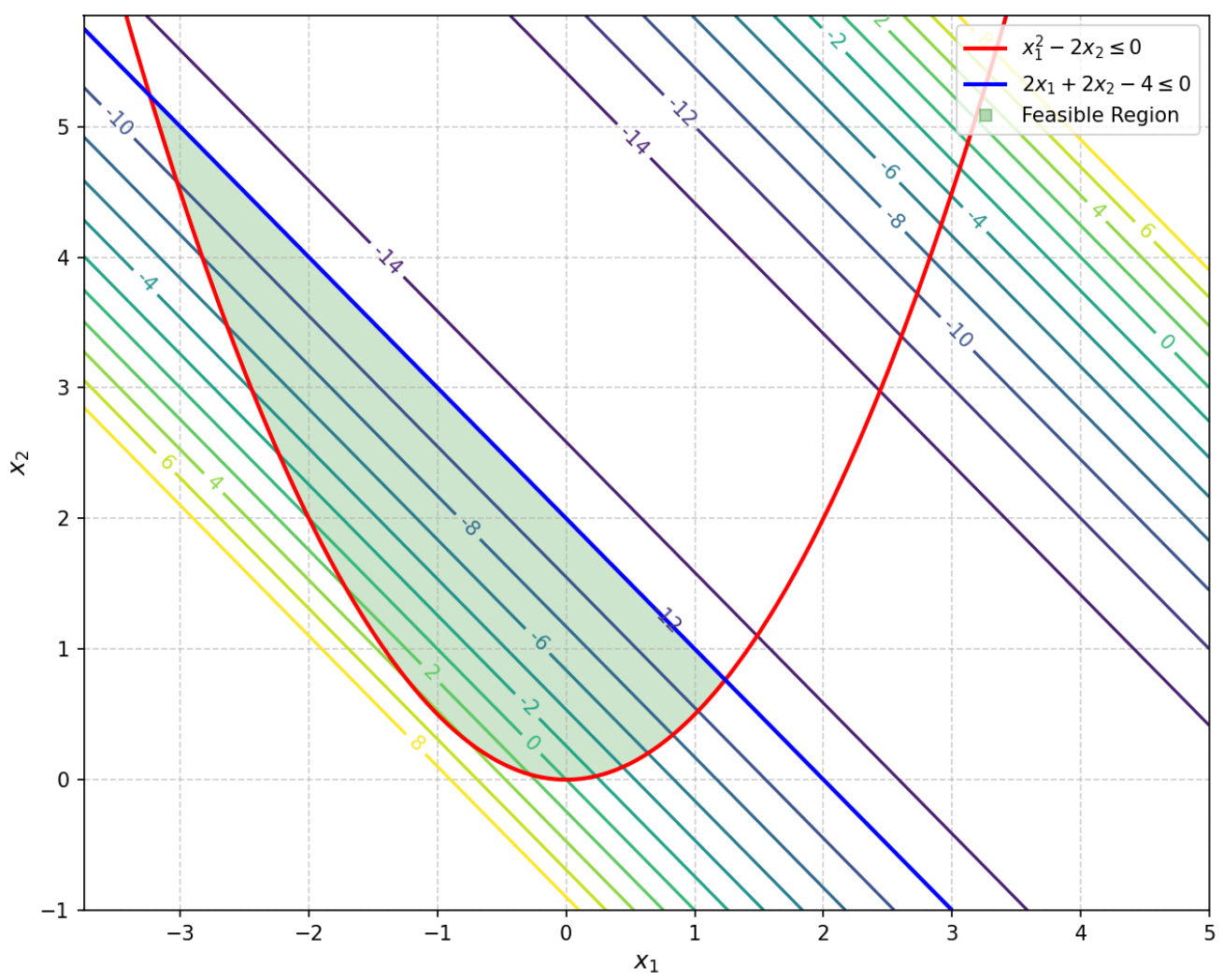
Recall that: $\nabla f(x^*)^T (y - x^*) = ||\nabla f(x^*)||\, ||y - x^*|| \cos(\theta)$

**$\nabla f(x^*)^T (y - x^*) \geq 0$ means that the angle between the vectors $\nabla f(x^*)$ and $y - x^*$ is always in $[0, \pi/2]$**

$$\min \quad f(x_1, x_2) = (x_1 + x_2)^2 - 8(x_1 + x_2)$$

$$s.t. \quad g_1(x_1, x_2) = x_1^2 - 2x_2 \leq 0$$

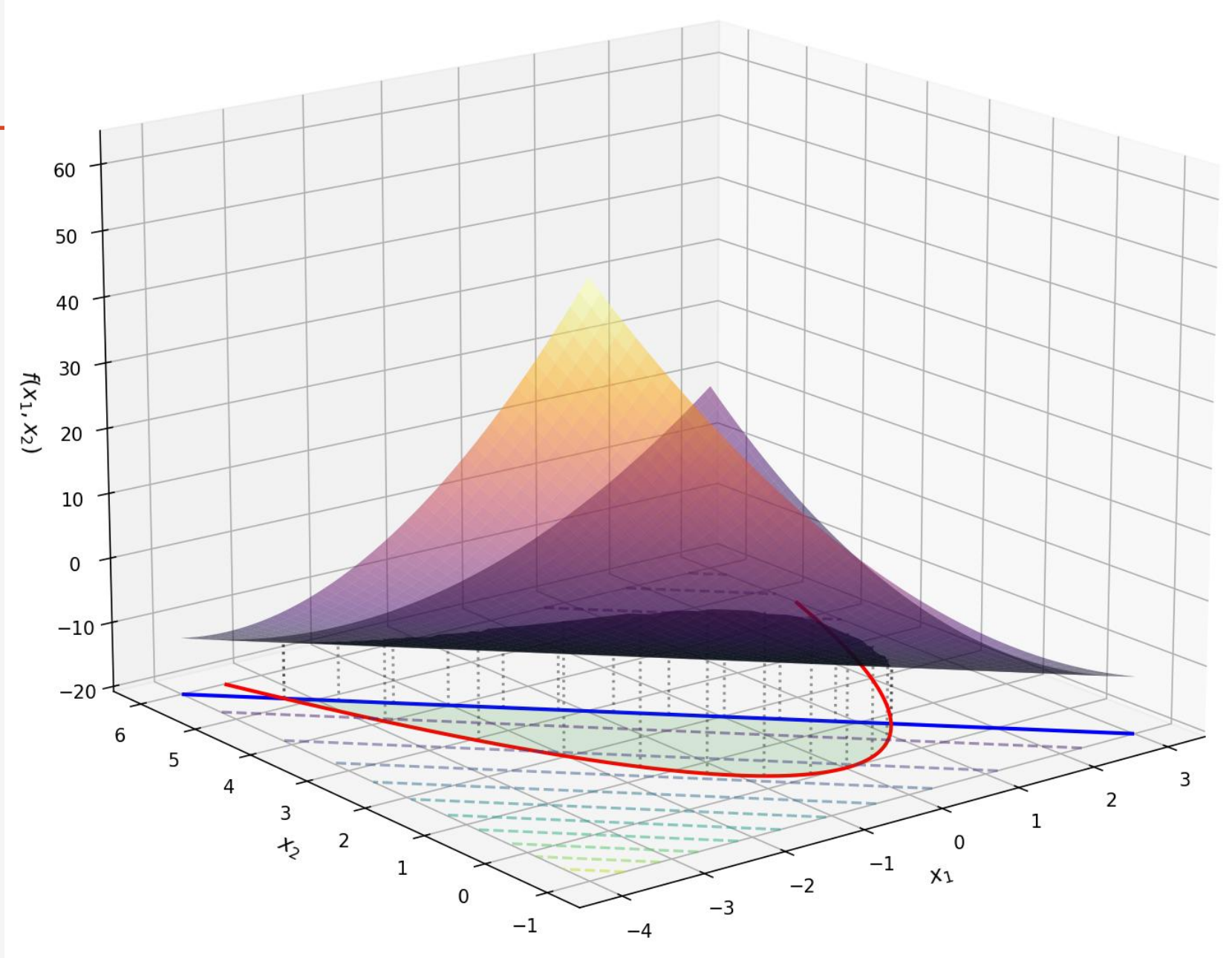
$$g_2(x_1, x_2) = 2x_1 + 2x_2 - 4 \leq 0$$



$$\min \quad f(x_1, x_2) = (x_1 + x_2)^2 - 8(x_1 + x_2)$$

$$s.t. \quad g_1(x_1, x_2) = x_1^2 - 2x_2 \leq 0$$

$$g_2(x_1, x_2) = 2x_1 + 2x_2 - 4 \leq 0$$



# Optimality Conditions for Constrained Convex Optimization

- **Proof**: We first prove: $x^* \in \Omega$ optimal $\Leftarrow$ $x^* \in \Omega$ and $\nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

# Optimality Conditions for Constrained Convex Optimization

- **Proof**: We first prove: $x^* \in \Omega$ optimal $\Leftarrow$ $x^* \in \Omega$ and $\nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

- Recall the 1st order characterization of convex functions:

$$\boxed{f(y) \geq \mathrm{f(x)} + \nabla f(x)^T(y - x), \forall x, y \in \Omega}$$

# Optimality Conditions for Constrained Convex Optimization

- **Proof**: We first prove: $x^* \in \Omega$ optimal $\Leftarrow$ $x^* \in \Omega$ and $\nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

- Recall the 1st order characterization of convex functions:

$$\boxed{f(y) \geq \mathrm{f(x)} + \nabla f(x)^T(y - x), \forall x, y \in \Omega}$$

- If we set $x = x^*$ we get: $f(y) \geq \mathrm{f(x^*)} + \nabla f(x^*)^T(y - x^*), \forall y \in \Omega$ (eq.1)

# Optimality Conditions for Constrained Convex Optimization

- **Proof**: We first prove: $x^* \in \Omega$ optimal $\Leftarrow$ $x^* \in \Omega$ and $\nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

- Recall the 1st order characterization of convex functions:

$$\boxed{f(y) \geq \mathrm{f(x)} + \nabla f(x)^T(y - x), \forall x, y \in \Omega}$$

- If we set $x = x^*$ we get: $f(y) \geq \mathrm{f(x^*)} + \nabla f(x^*)^T(y - x^*), \forall y \in \Omega$ (eq.1)

- Since $\nabla f(x^*)^T(y - x^*) \geq 0$, we can write (eq.1) as follows:

$$f(y) \geq \mathrm{f(x^*)} + \nabla f(x^*)^T(y - x^*) \geq f(x^*), \forall y \in \Omega \;\Rightarrow\; \mathrm{f(y)} \geq f(x^*), \forall y \in \Omega$$

Hence $x^*$ is optimal in $\Omega$

**Note**: we have not used FONC!

# Optimality Conditions for Constrained Convex Optimization

- We will now prove: $x^* \in \Omega$ optimal $\Rightarrow \nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

# Optimality Conditions for Constrained Convex Optimization

- We will now prove: $x^* \in \Omega$ optimal $\Rightarrow \nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

- We will use proof by contradiction. Assume that $\exists y \in \Omega \; st \; \nabla f(x^*)^T(y - x^*) < 0$

# Optimality Conditions for Constrained Convex Optimization

- We will now prove: $x^* \in \Omega$ optimal $\Rightarrow \nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$
- We will use proof by contradiction. Assume that $\exists y \in \Omega \; st \; \nabla f(x^*)^T(y - x^*) < 0$
- Consider the univariate function: $g(\alpha) = f\big(x^* + \alpha(y - x^*)\big)$, for $\alpha > 0$.

# Optimality Conditions for Constrained Convex Optimization

- We will now prove: $x^* \in \Omega$ optimal $\Rightarrow \nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$
- We will use proof by contradiction. Assume that $\exists y \in \Omega \; st \; \nabla f(x^*)^T(y - x^*) < 0$
- Consider the univariate function: $g(\alpha) = f\big(x^* + \alpha(y - x^*)\big)$, for $\alpha > 0$.
- Take the derivative: $g'(\alpha) = \nabla f\big(x^* + \alpha(y - x^*)\big)^T(y - x^*)$ (→ chain rule)
- Setting $\alpha = 0$ and using our assumption: $g'(0) = \nabla f(x^*)^T(y - x^*) < 0$ (from our assumption we have $< 0$ )

# Optimality Conditions for Constrained Convex Optimization

- We will now prove: $x^* \in \Omega$ optimal $\Rightarrow \nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in \Omega$

- We will use proof by contradiction. Assume that $\exists y \in \Omega \; st \; \nabla f(x^*)^T(y - x^*) < 0$

- Consider the univariate function: $g(\alpha) = f\big(x^* + \alpha(y - x^*)\big)$, for $\alpha > 0$.

- Take the derivative: $g'(\alpha) = \nabla f\big(x^* + \alpha(y - x^*)\big)^T(y - x^*)$ (→ chain rule)

- Setting $\alpha = 0$ and using our assumption: $g'(0) = \nabla f(x^*)^T(y - x^*) < 0$ (from our assumption we have $< 0$ )

- From the definition of the derivative: $\lim_{\alpha \to 0} \frac{g(\alpha) - g(0)}{\alpha} < 0$, which means that

$$\exists \delta > 0, \; s.t. \; \frac{g(\alpha) - g(0)}{\alpha} < 0, \; \forall \alpha \in (0, \delta) \Rightarrow g(\alpha) < g(0), \; \forall \alpha \in (0, \delta) \overset{(\alpha > 0)}{\Rightarrow} f\big(x^* + \alpha(y - x^*)\big) < f(x^*), \; \forall \alpha \in (0, \delta) \quad \text{(eq.2)}$$

- But (eq.2) **contradicts** the fact that $x^*$ is optimal. Hence our assumption that $\exists y \in \Omega \; st \; \nabla f(x^*)^T(y - x^*) < 0$ does not hold. ■

# CVX demo

$$\begin{aligned} &min && f(x) \\ &s.t. && g_i(x) \le 0, i = 1, \dots, m \\ & && h_j(x) = 0, j = 1, \dots, r \end{aligned}$$

$$\begin{aligned} &min && f(x) \\ &s.t. && g(x) \le 0 \\ & && h(x) = 0 \end{aligned}$$

$$g(x) = \left(g_1(x), \dots, g_m(x)\right)^T, \qquad g: R^n \to R^m$$

$$h(x) = (h_1(x), \dots, h_r(x))^T, \qquad h: R^n \to R^r$$

$$h_j(x) := a_j^T x - b_j = 0, j = 1, \dots, r \quad \text{(affine functions)}$$

$$h(x) = Ax - b$$

```
cvx_begin
      variables x(n)
      minimize(f(x))
      subject to
            g(x) <= 0
            A*x - b == 0
cvx_end
```

After termination CVX returns the following information:

- $x$ → solution vector ($x^*$)
- cvx_optval → the optimal value ($f(x^*)$)
- cvx_status → solver status (e.g., Solved, Unbounded, Infeasible, …)
- For more information check the documentation!

# CVX demo

```
cvx_begin

      Variables x1 x2

      Maximize(140*x1 + 235*x2)

      Subject to

              x1 + x2 <= 180

              x1 + 2*x2 <= 240

              0.3*x1 + 0.1*x2 <= 30

              x1 >=0

              x2 >= 0

cvx_end

cvx_optimal % to see the optimal value
```

```
% least squares example

m = 16; n = 8;

A = randn(m, n); b = randn(m,1)

cvx_begin

      Variable x(n)

      minimize(norm(A*x-b))

      % maximize(norm(A*x-b))

cvx_end
```

**Resources**: http://cvxr.com/cvx/doc/quickstart.html

# CVX examples

```
cvx_begin
        variables x(1), y(1)
        minimize(x + y)
        subject to
                x >= 1
                y == 2
cvx_end
```

Upon termination of the optimization algorithm, CVX returns:

```
>> x
        1
>> y
        2
>> cvx_optimal
        3
>> cvx_status
        Solved
```

```
General formulation of a Linear
Programming problem
(Note: you must define n, A, c
and b before cvx_begin)

cvx_begin
        variables x(n)
        maximize(c'*x)
        subject to
                A*x == b
                x >= 0
cvx_end
```

# CVXPY → CVX for Python

**You can find excellent resources at:**

https://www.cvxpy.org/

**A lot of examples at:**

https://www.cvxpy.org/examples/index.html

CVXPY example: Constrained Least Squares

```python
import cvxpy as cp
import numpy as np

# Problem data.
m = 30
n = 20
np.random.seed(1)
A = np.random.randn(m, n)
b = np.random.randn(m)

# Construct the problem.
x = cp.Variable(n)
objective = cp.Minimize(cp.sum_squares(A @ x - b))
constraints = [0 <= x, x <= 1]
prob = cp.Problem(objective, constraints)

# The optimal objective value is returned by `prob.solve()`.
result = prob.solve()
# The optimal value for x is stored in `x.value`.
print(x.value)
# The optimal Lagrange multiplier for a constraint is stored in
# `constraint.dual_value`.
print(constraints[0].dual_value)
```

# Proofs of the 1st and 2nd order characterization of convex functions

# 1st and 2nd order characterization of convex functions

- **Theorem 2**: suppose $f: R^n \to R$ is twice continuously differentiable. Then the following statements are equivalent

Definition of convexity – no derivatives needed

- **0th order:** $f$ is convex $\left(f(\lambda x + (1-\lambda)y \leq \lambda f(x) + (1-\lambda)f(y), \boxed{\forall x, y \in R^n, \lambda \in [0,1]}\right)$

$$\Updownarrow$$

Does it remind you anything??

- **1st order:** $f(y) \geq f(x) + \nabla f(x)^T (y - x), \boxed{\forall x, y \in R^n}$ 1st order derivatives needed

$$\Updownarrow$$

- **2nd order:** $\nabla^2 f(x) \succcurlyeq 0, \boxed{\forall x \in R^n}$ 2nd order derivatives needed

- **Proof of:** $f$ is convex $\Leftrightarrow$ $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- ($\Rightarrow$) since $f$ is convex: $\forall x, y$ and $\lambda \in [0,1]$ : $f\big((1-\lambda)x + \lambda y\big) \leq (1-\lambda) f(x) + \lambda f(y) \Rightarrow$

- **Proof of:** $f$ is convex $\iff$ $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- ($\Rightarrow$) since $f$ is convex: $\forall x, y$ and $\lambda \in [0,1]$ : $f\big((1-\lambda)x + \lambda y\big) \leq (1-\lambda) f(x) + \lambda f(y) \Rightarrow$

$$\Rightarrow f\big(x + \lambda(y - x)\big) \leq f(x) + \lambda\big(f(y) - f(x)\big) \Rightarrow f\big(x + \lambda(y - x)\big) - f(x) \leq \lambda\big(f(y) - f(x)\big) \Rightarrow$$

- **Proof of:** $f$ is convex $\iff$ $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- ($\Rightarrow$) since $f$ is convex: $\forall x, y$ and $\lambda \in [0,1]$ : $f\big((1-\lambda)x + \lambda y\big) \leq (1-\lambda) f(x) + \lambda f(y) \Rightarrow$

$$\Rightarrow f\big(x + \lambda(y - x)\big) \leq f(x) + \lambda\big(f(y) - f(x)\big) \Rightarrow f\big(x + \lambda(y - x)\big) - f(x) \leq \lambda\big(f(y) - f(x)\big) \Rightarrow$$

$$\Rightarrow \frac{f\big(x + \lambda(y - x)\big) - f(x)}{\lambda} \leq f(y) - f(x) \Rightarrow f(y) \geq f(x) + \frac{f\big(x + \lambda(y - x)\big) - f(x)}{\lambda}$$

- **Proof of:** $f$ is convex $\iff$ $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- ($\Rightarrow$) since $f$ is convex: $\forall x, y$ and $\lambda \in [0,1]$ : $f\big((1-\lambda)x + \lambda y\big) \leq (1-\lambda) f(x) + \lambda f(y) \Rightarrow$

$$\Rightarrow f\big(x + \lambda(y - x)\big) \leq f(x) + \lambda\big(f(y) - f(x)\big) \Rightarrow f\big(x + \lambda(y - x)\big) - f(x) \leq \lambda\big(f(y) - f(x)\big) \Rightarrow$$

$$\Rightarrow \frac{f\big(x + \lambda(y - x)\big) - f(x)}{\lambda} \leq f(y) - f(x) \Rightarrow f(y) \geq f(x) + \frac{f\big(x + \lambda(y - x)\big) - f(x)}{\lambda}$$

Taking limits as $\lambda \to 0$:

$$f(y) \geq f(x) + \lim_{\lambda \to 0} \frac{f\big(x + \lambda(y - x)\big) - f(x)}{\lambda} \Rightarrow f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

- ($\Leftarrow$) we know $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- ($\Leftarrow$) we know $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- Define $z = \lambda x + (1 - \lambda) y$ and apply the 1st order characterization twice

$$f(x) \geq f(z) + \nabla f(z)^T (x - z) \ , \qquad f(y) \geq f(z) + \nabla f(z)^T (y - z)$$

- ($\Leftarrow$) we know $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- Define $z = \lambda x + (1 - \lambda) y$ and apply the 1st order characterization twice

$$f(x) \geq f(z) + \nabla f(z)^T (x - z) \ , \qquad f(y) \geq f(z) + \nabla f(z)^T (y - z)$$

- Multiplying the first by $\lambda$ and the second by $(1 - \lambda)$ and add them up yields

$$\lambda f(x) + (1 - \lambda) f(y) \geq \lambda f(z) + \lambda \nabla f(z)^T (x - z) + (1 - \lambda) f(z) + (1 - \lambda) \nabla f(z)^T (y - z) =$$

- ($\Leftarrow$) we know $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n$

- Define $z = \lambda x + (1 - \lambda) y$ and apply the 1st order characterization twice

$$f(x) \geq f(z) + \nabla f(z)^T (x - z) \ , \qquad f(y) \geq f(z) + \nabla f(z)^T (y - z)$$

- Multiplying the first by $\lambda$ and the second by $(1 - \lambda)$ and add them up yields

$$\lambda f(x) + (1 - \lambda) f(y) \geq \lambda f(z) + \lambda \nabla f(z)^T (x - z) + (1 - \lambda) f(z) + (1 - \lambda) \nabla f(z)^T (y - z) =$$

$$= f(z) + \nabla f(z)^T (\lambda x + (1 - \lambda) y - z) = f(\lambda x + (1 - \lambda) y) + 0 \Rightarrow$$

$$\Rightarrow f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y) \qquad \blacksquare$$

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \quad \forall x \in R^n$

- We show first the case where $\boldsymbol{n = 1}$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T (y - x), \;\; \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \;\; \forall x \in R^n$

- We show first the case where $n = 1$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

$$f(y) \geq f(x) + f'(x)(y - x) \;\; and \quad f(x) \geq f(y) + f'(y)(x - y) \Rightarrow$$

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T (y - x), \;\; \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \;\; \forall x \in R^n$

- We show first the case where $n = 1$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

$$f(y) \geq f(x) + f'(x)(y - x) \;\; and \quad f(x) \geq f(y) + f'(y)(x - y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y - x) \;\; and \;\; f(x) - f(y) \geq f'(y)(x - y) \Rightarrow$$

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T(y-x), \;\; \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \;\; \forall x \in R^n$

- We show first the case where $n = 1$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

$$f(y) \geq f(x) + f'(x)(y-x) \;\; and \quad f(x) \geq f(y) + f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(x) - f(y) \geq f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(y) - f(x) \leq f'(y)(y-x) \Rightarrow$$

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T(y-x), \;\; \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \;\; \forall x \in R^n$

- We show first the case where $n = 1$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

$$f(y) \geq f(x) + f'(x)(y-x) \;\; and \quad f(x) \geq f(y) + f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(x) - f(y) \geq f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(y) - f(x) \leq f'(y)(y-x) \Rightarrow$$

$$\Rightarrow f'(x)(y-x) \leq f(y) - f(x) \leq f'(y)(y-x) \Rightarrow$$

- **Proof of:** $f(y) \geq f(x) + \nabla f(x)^T(y-x), \;\; \forall x, y \in R^n \quad \Leftrightarrow \quad \nabla^2 f(x) \succcurlyeq 0, \;\; \forall x \in R^n$

- We show first the case where $n = 1$

- ($\Rightarrow$) we apply the inequality for $x$ and $y$ separatel

$$f(y) \geq f(x) + f'(x)(y-x) \;\; and \quad f(x) \geq f(y) + f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(x) - f(y) \geq f'(y)(x-y) \Rightarrow$$

$$\Rightarrow f(y) - f(x) \geq f'(x)(y-x) \;\; and \;\; f(y) - f(x) \leq f'(y)(y-x) \Rightarrow$$

$$\Rightarrow f'(x)(y-x) \leq f(y) - f(x) \leq f'(y)(y-x) \Rightarrow$$

$$f'(x)(y-x) \leq f'(y)(y-x) \Rightarrow \big(f'(y) - f'(x)\big)(y-x) \geq 0$$

- We divide now the last inequality by $(y-x)^2$ , when $y \neq x$, and get

$$\frac{\big(f'(y)-f'(x)\big)(y-x) \geq 0}{(y-x)^2} \geq 0 \Rightarrow \frac{f'(y)-f'(x)}{(y-x)} \geq 0$$

- Taking limit as $y \to x$ yields:

$$\lim_{y \to x} \frac{f'(y)-f'(x)}{(y-x)} \geq 0 \Rightarrow f''(x) \geq 0$$

- ($\Leftarrow$) we now know that $f''(x) \geq 0, \forall x \in domf$.

- We take the **2nd order Taylor expansion** of $f$ at $x$

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2} f''(z)(y - x)^2$$

for some $z \in domf$.

- Since $f''(x) \geq 0, \forall x \in domf$ we have: $f(y) \geq f(x) + f'(x)(y - x)$

- **Now we will prove the general case when $n \geq 2$**

- We know that $f$ is convex function if its restriction along all lines is convex, that is

$$g(t) = f(x + td)$$

is convex function for all $x \in domf$, $d \in R^n$ and $t: x + td \in domf$.

- But $g(t)$ is a univariate convex function. Hence $g''(t) \geq 0$.
  - $g'(t) = \nabla f(x + td)^T d$ and $g''(x) = d^T \nabla^2 f(x + td) d \geq 0, \forall x \in domf,\ d \in R^n, x + td \in domf$
- Hence $\nabla^2 f(x) \succcurlyeq 0, \forall x \in domf$

$\blacksquare$